

# Hamza Mustafa

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## EDUCATION

### Skidmore College

*Bachelor of Arts in Computer Science, Minor in Mathematics, Major GPA: 3.95*

Saratoga Springs, NY

*Aug. 2023 – May 2027*

## EXPERIENCE

### Bloomberg Terminal, Quantative Researcher

*Skidmore College*

May 2026 – August 2026

*Saratoga Springs, NY*

- Trained PPO models on SPY asset data that successfully **outperformed** a standard buy and hold agent by **20%**
- Tested whether hourly vs daily data performed better at trading options in the **P&L evaluation**
- Presented “Can AI learn to trade options without seeing options” to **120+ attendees**

### Backend Software Engineer

*ProgressSoft Corporation*

July 2025 – September 2025

*Amman, Jordan*

- Designed and developed scalable **RESTful backend services** using **Spring Boot** and relational databases (**PostgreSQL**, **MySQL**) to support secure financial transactions and ensure data integrity and compliance
- Built and validated APIs through **unit and integration testing**, improving reliability
- Secured 8+ REST API endpoints via JWT-based authentication and role-based authorization, reducing unauthorized access risk while optimizing query and service logic to cut API latency by 25%
- Containerized services with **Docker** to enable scalable deployment and cross-team integration of web-based tools

### Nvidia IsaacLab, Reinforcement Learning Research Assistant

*Skidmore College*

Sep 2024 – August 2025

*Saratoga Springs, NY*

- Engineered reinforcement learning pipelines using **NVIDIA Isaac Lab** to achieve stable locomotion using PID controller, **improving gait stability** in response to outside forces by **40%** over baseline controllers .
- Designed and tuned reward functions, simulation parameters, and control policies, **reducing training time by 30%** while increasing convergence reliability over 4096 parallel environments.
- Presented “Bipedal Locomotion in Quadrupedal Robots Using Reinforcement Learning” to 120+ attendees

## PROJECTS

### ThriftBred | *SpringBoot, NextJs, PostgreSQL, Docker*

Nov. 2025 — Present

- Built a full-stack marketplace application for Skidmore College using Spring Boot, Next.js, and PostgreSQL, supporting authenticated users, item listings, and peer-to-peer purchases with a scalable REST-based architecture
- Implemented secure user authentication and integrated Stripe payments for end-to-end checkout flows, ensuring safe transaction handling and reliable data persistence across users, listings, and orders.

### Shift2Sakura | *NodeJS, NextJS, PostgreSQL*

January 2026 – April 2026

- Engineered a full-stack AI-powered resume conversion platform using Next.js, Express.js, and PostgreSQL, enabling non-Japanese applicants to generate Japan-standard Rirekisho PDFs through a multi-step conversion pipeline
- Integrated a locally-hosted LLaMA 3 model via Ollama to extract, translate, and restructure English resume content into Japanese hiring format, reducing manual formatting effort by processing resumes in under 60 seconds
- Implemented JWT-based authentication using Clerk across both frontend and backend, securing API endpoints with middleware and persisting user-specific resume conversion history in a PostgreSQL database via Prisma ORM
- Designed a PostgreSQL relational database schema using Prisma ORM, implementing migration workflows and role-scoped data access to ensure each authenticated user could retrieve and download only their conversion history.

## TECHNICAL SKILLS

**Languages:** Java, Python, C, C# (Unity), SQL (PostgreSQL, MySQL), JavaScript, TypeScript, HTML/CSS, R

**Frameworks & Technologies:** Spring Boot, Next.js, React, REST APIs, Docker, CI/CD, NVIDIA Isaac Lab

**Databases:** PostgreSQL, MySQL, relational database design and optimization

**Cloud & Infrastructure:** AWS (familiar), containerized deployment, distributed systems

**Tools and Practices:** Git, GitHub, Agile/Scrum, code review, unit & integration testing

**Libraries:** PyTorch, pandas, NumPy, Matplotlib